Muhammad Qois Huzyan Octava

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About Me

Final-year Master's student in Information Engineering at Universitas Gadjah Mada, with a strong enthusiasm for data-driven technologies. Experienced as a teaching and research assistant in data mining, contributing to data analysis, experimentation, and research support using Python and SQL. Previously worked as a system analyst, focusing on data modeling, data quality, and performance optimization in PostgreSQL.

Education

Universitas Gadjah Mada, M.Eng. in Information Engineering — February 2024 – July 2026

- Current GPA: 3.5/4.0
- Currently in the final stage of completing Master's thesis on data engineering pipelines and preprocessing strategies to improve domain-specific speech recognition performance.

Universitas Gadjah Mada, S.Tr.Kom. in Software Engineering Technology — August 2019 – July 2023

- Graduate GPA: 3.7/4.0
- Conducted applied research in Machine Learning, Web Development, and Natural Language Processing, bridging academic concepts with real-world applications.

Professional Training

Dibimbing, Data Engineering Bootcamp — January 2026 – July 2026

- Training in data engineering practices through a 6-month intensive program covering ETL pipeline development and data warehouse implementation using Python, PostgreSQL, Airflow, dbt, Docker, and AWS/GCP.

Experience

System Analyst for Database at FMIPA, Universitas Gadjah Mada – Yogyakarta — May 2024 – December 2024

- Performed database structure assessment and optimization as part of the Master Data Management (MDM) team for the Minerbaone project under Indonesia's Ministry of Energy and Mineral Resources (ESDM).
- Designed and developed unified database architecture, following a comprehensive analysis of thousands of tables across multiple existing systems and regulatory frameworks to establish a single source.
- Facilitated cross-functional collaboration with engineering teams to ensure the new database system met all technical and business requirements.

Teaching Assistant at Sekolah Vokasi, Universitas Gadjah Mada – Yogyakarta — January 2023 – June 2025

- Facilitated hands-on laboratory sessions for data-related courses, including Artificial Intelligence (2023), Data Mining (2024; 2025), and Distributed Administration and Information Systems (2024), enabling students to apply theoretical concepts in practical settings.
- Guided students in implementing data mining and machine learning techniques using Python, applying analytical methods to real-world datasets.
- Mentored students in troubleshooting programming tasks and understanding database fundamentals and distributed system environments, while evaluating assignments and examinations aligned with course objectives.

Projects

Contributed for LLM-RAG System for Disaster Mitigation Project — 2025

- Fine-tuned Whisper model for domain specific needs with word error rate optimization up to 2.5x of vanilla model, HuggingFace: octava/whisper-small-disaster
- Collected data needed for RAG by transcribing audio from YouTube videos into structured text data and Automated relevant text data filtration by utilizing LLM models (chatGPT and Gemini API).

MinerbaOne: Indonesian Mineral and Coal Mining Governance Web App 2024

- Designed and developed comprehensive Entity Relationship Diagrams (ERD) by translating complex regulatory requirements and existing systems into scalable and well-structured database models.
- Designed a dynamic data warehouse architecture to support historical data tracking and enable future analytical and reporting use cases.
- Optimized ORM queries by analyzing and refactoring them into raw PostgreSQL queries.

Web-Based Sentiment Analysis Dashboard 2023

- Developed and deployed a sentiment analysis model using TF-IDF enhanced with statistical feature engineering and Support Vector Machine (SVM).
- Implemented the model into a web-based dashboard that enables intuitive visualization and sentiment analysis.
- The project findings were presented at the 3ICT conference and have been published in the IEEE proceedings.

Technical Skills

Programming Languages: Proficient in Python, JavaScript, and SQL for AI and Software Development.

Data and AI Technologies: Experienced with Machine Learning workflows using Scikit-learn, Pandas, and NumPy; capable of building intelligent systems with Pytorch and Transformers.

Web Backend Development: Capable of delivering APIs and web applications using FastAPI, Flask, ExpressJS, and NestJS. Experienced in managing and optimizing relational databases using MySQL and PostgreSQL.

Certifications

Microsoft Certified: Azure AI Engineer Associate, Microsoft March 2024

- Credential ID: F460DFBCE97A67A6

Huawei Certified ICT Associate - AI, Huawei October 2021

- Credential ID: 010102001397808454091226803

Organizational Experience

Unit Coordinator, KKN-PPM Universitas Gadjah Mada June 2022 – August 2022

As the leader of unit 2022-YO042, directed a multidisciplinary team of 26 students in a community service program focused on environmental and waste management initiatives. Coordinated project planning, implementation, and evaluation across education, technology, and sustainability initiatives, fostering collaboration between university and local communities.

Publications

Fine-Tuning Whisper for Domain-Specific ASR: Transcribing Indonesian YouTube Content on Local Wisdom in Disaster Mitigation September 2025

Octava, M.Q.H., Pratama, A.R., Alfarozi, S.A.I.

DOI: 10.1109/ICAICTA67604.2025.11335113

Application of the outlier detection method for web-based blood glucose level monitoring system August 2024

Nurhaliza, R.A., ***Octava, M.Q.H.***, Hilmy, F.M., Farooq, U., Alfian, G.

DOI: 10.11591/eei.v13i4.7717

Web-based Sentiment Analysis System Using SVM and TF-IDF with Statistical Feature November 2023

Octava, M.Q.H., Putri, D.G.P., Hilmy, F.M., Farooq, U., Nurhaliza, R.A., Alfian, G.

DOI: 10.1109/3ICT60104.2023.10391734